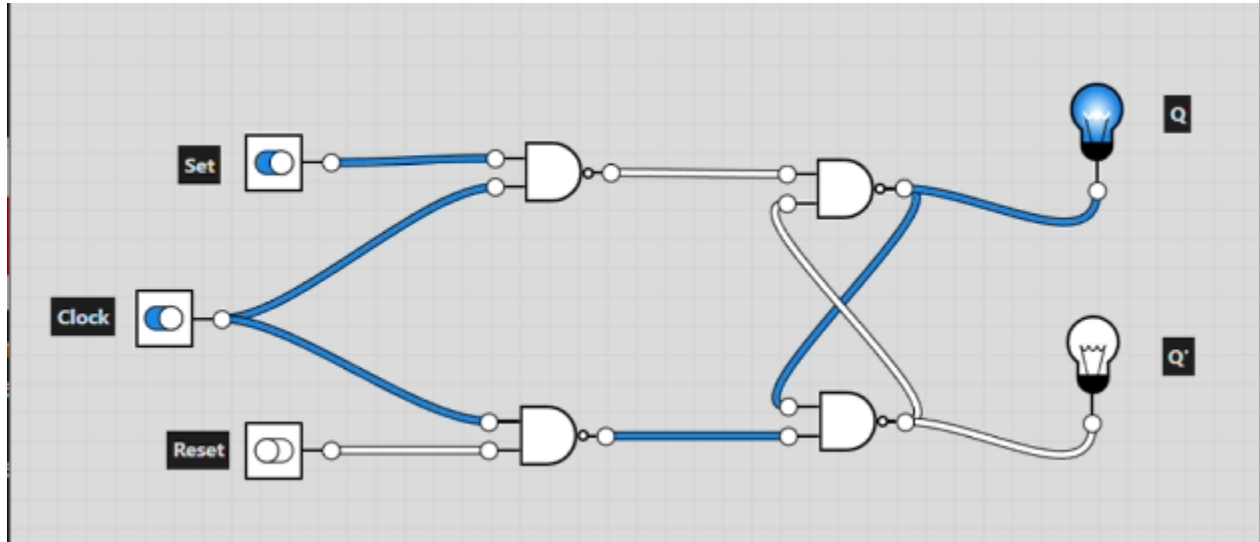


Experiment No.11 APPLICATION OF SR FLIP-FLOP	
Course Code: CPE 013	Program: BSCpE
Course Title: Logic Circuits and Design	Date Performed: 14/4/2026
Section: CPE22S2	Date Submitted: 14/4/2026
Student Name: Bulaong, Ross Andrew D. Florencio, David Malabanan, Nadine Que, Leroy Josef Zapanta, Russell	Instructor: Engr. Noel Lusung
1. Instructions	
Research about an application of SR flipflop and simulate. The simulation must be in the tinkercad with a file name of: EXPERIMENT11_GROUPNUMBER_FLIPFLOP Submit a document containing the following: ● Description of your work ● Screenshot of the diagram ● Screenshot of the code ● Direct link of the tinkercad ● Conclusion ● Learnings to the activity ● What are the participation of every group members? Kindly indicate NONE if they did not participate in the laboratory. ● This experiment will also served as your QUIZ. ● Only 1 representative from the group will submit the whole document	

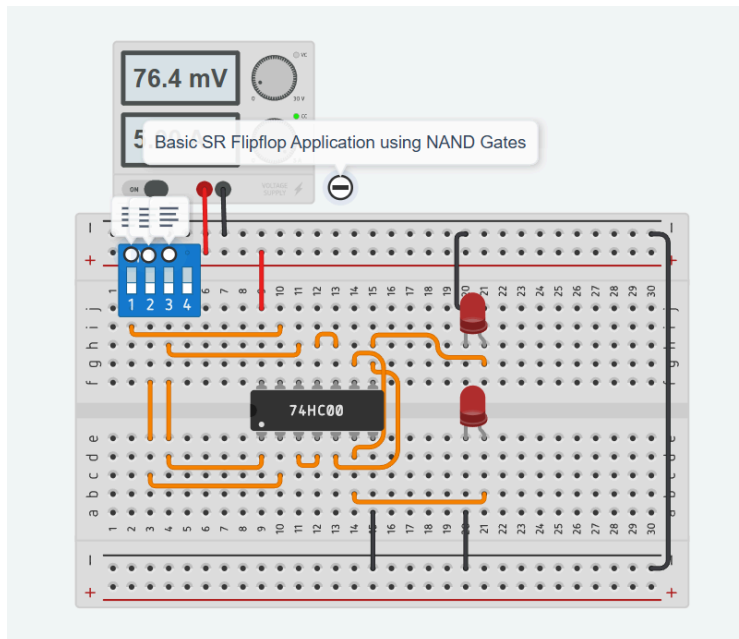
Basic SR Latch Circuit using NAND Gates

Logicly Circuit Config



Video Demo:  **Logicly**

TinkerCAD Circuit Config



Video Demo: **Tinkercad**

Truth Table / Input Table

Clock	Set	Reset	Q(next)	Q'(next)
0	X	X	Q(no change)	Q'(no change)
1	0	0	1	1
1	0	1	1	0
1	1	0	0	1
1	1	1	Q(no change)	Q'(no change)

Conclusion

As a group, we successfully designed and simulated an SR flip-flop using NAND gates in both Logically and TinkerCAD.

The experiment helped us understand how the SR flip-flop works as a basic memory element in digital circuits.

We observed how the Set and Reset inputs affect the output Q and its complement Q'.

The truth table we created matched the actual behavior of the circuit during simulation.

We also learned how the circuit maintains its state when both inputs are inactive.

The role of the clock signal in controlling when changes occur was clearly demonstrated.

We recognized the importance of avoiding invalid input conditions to prevent unpredictable outputs.

This activity improved our skills in circuit design and simulation tools. We became more confident to create and build circuits, testing different scenarios, and troubleshooting errors. It also enhanced our ability to use simulation software effectively as part of the design process.

It also strengthened our understanding of sequential logic circuits. Overall this activity connected theoretical concepts with hands-on experience making it easier for us to understand how digital systems operate in real world applications.

Contribution Chart

Name	Contribution
Bulaong, Ross Andrew D.	TinkerCAD simulation
Florencio, David	Truth table

Malabanan, Nadine	Truth table
Que, Leroy Josef	Logicly simulation
Zapanta, Russell	Truth table